

Relativistic kinematics

For Parts 1 and 2, let B's inertial frame S' move at velocity $+v$ along the x -axis of A's frame S . With parallel axes and coincident origins at $t = t' = 0$, the Lorentz transformations are

$$x' = \gamma(x - vt), \quad t' = \gamma\left(t - \frac{vx}{c^2}\right),$$

and their inverses are

$$x = \gamma(x' + vt'), \quad t = \gamma\left(t' + \frac{vx'}{c^2}\right), \quad \gamma = \frac{1}{\sqrt{1 - v^2/c^2}}.$$

Coordinate differences between any two events obey the same transformations.

1. Time dilation

B carries a clock stationary in S' . Consider two successive ticks. Both occur at the same position in B's frame, so

$$\Delta x' = 0.$$

The interval recorded by this clock is the *proper time*, $\Delta\tau = \Delta t'$. Applying the inverse time transformation to the two ticks gives

$$\Delta t = \gamma\left(\Delta t' + \frac{v\Delta x'}{c^2}\right) = \gamma\Delta t'.$$

Therefore

$$\boxed{\Delta t = \gamma\Delta\tau.}$$

For $v \neq 0$, $\gamma > 1$: A assigns a longer interval between the ticks than B's clock records. During an interval Δt in A's frame, the moving clock advances by $\Delta t/\gamma$. The interval Δt is determined using synchronised clocks in A's frame at the tick locations.

2. Length contraction

B holds a rod stationary in S' , parallel to the x' -axis. Its length in its rest frame is the *proper length*, $L_0 = x'_2 - x'_1$.

To measure its length in S , A records the positions of both ends *simultaneously in A's frame*. These two measurement events satisfy

$$\Delta t = 0, \quad L = \Delta x.$$

The spatial transformation gives

$$\Delta x' = \gamma(\Delta x - v\Delta t) = \gamma\Delta x.$$

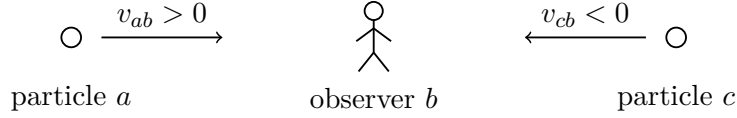
The endpoints have fixed coordinates in S' , so $\Delta x' = L_0$, even though these events are not simultaneous in S' . Thus

$$\boxed{L = \frac{L_0}{\gamma} = L_0\sqrt{1 - \frac{v^2}{c^2}}.}$$

For these same events, $\Delta t' = -\gamma vL/c^2$. One cannot also impose $\Delta t' = 0$; the length measurement requires simultaneity in the measuring frame.

3. Velocity addition

Take rightwards as positive. Particle a is on the left, observer b is in the middle, and particle c is on the right. Both particles move towards b , as shown in b 's rest frame.



Define v_{ij} as the signed velocity of i measured in the rest frame of j . In particular,

$$\begin{aligned} v_{ab} &= \text{velocity of } a \text{ relative to } b, \\ v_{bc} &= \text{velocity of } b \text{ relative to } c, \\ v_{ac} &= \text{velocity of } a \text{ relative to } c. \end{aligned}$$

Here $v_{bc} = -v_{cb} > 0$. The leftward arrow in the diagram represents v_{cb} , whereas the addition formula below uses v_{bc} . The unsubscripted c in the equations denotes the speed of light.

Use (x_b, t_b) for coordinates in b 's rest frame and (x_c, t_c) for coordinates in particle c 's rest frame. Since b 's frame moves at $+v_{bc}$ relative to c 's frame, the Lorentz transformations give

$$x_c = \gamma_{bc}(x_b + v_{bc}t_b), \quad t_c = \gamma_{bc} \left(t_b + \frac{v_{bc}x_b}{c^2} \right),$$

where $\gamma_{bc} = 1/\sqrt{1 - v_{bc}^2/c^2}$ and the coordinate origins are chosen to coincide at a common event.

For two infinitesimally separated events *on particle a's trajectory*,

$$dx_c = \gamma_{bc}(dx_b + v_{bc}dt_b), \quad dt_c = \gamma_{bc} \left(dt_b + \frac{v_{bc}dx_b}{c^2} \right).$$

By definition,

$$v_{ab} = \frac{dx_b}{dt_b}, \quad v_{ac} = \frac{dx_c}{dt_c}.$$

Dividing the transformed intervals,

$$v_{ac} = \frac{dx_b + v_{bc}dt_b}{dt_b + v_{bc}dx_b/c^2} = \frac{dx_b/dt_b + v_{bc}}{1 + (v_{bc}/c^2)(dx_b/dt_b)}.$$

Hence the one-dimensional velocity addition equation is

$$\boxed{v_{ac} = \frac{v_{ab} + v_{bc}}{1 + v_{ab}v_{bc}/c^2}.$$